FIITJEE **REVIEW TEST - 04**

PHYSICS, CHEMISTRY & MATHEMATICS

REVIEW TEST

Time Allotted: 3 Hours **Maximum Marks: 198**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **SECTION–I** is Physics, **SECTION–II** is Chemistry and **SECTION–III** is Mathematics.
4. Each part is further divided into **two parts**: **Part – A & C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

(i) **Part–A (01 – 06)** contains 06 multiple choice questions which have one or more than one correct answer. Each question will be evaluated according to the following marking scheme.

Full Marks	:	**+4**	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	**+3**	If all the four options are correct but ONLY three options are chosen;
Partial Marks	:	**+2**	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	**+1**	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	**0**	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	**–1**	In all other cases.

(ii) **Part–C (07 – 12)** contains 06 Numerical based questions with single digit integer as answer, ranging from 0 to 9 (both inclusive) and each question carries **+3 marks** for correct answer and **0 mark** for wrong answer.

(iii) **PART–C (13 – 18)** contains 06 Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer. **There is no negative marking.**

Name of the Candidate :_____

Batch :_____ **Date of Examination :**_____

Enrolment Number :_____

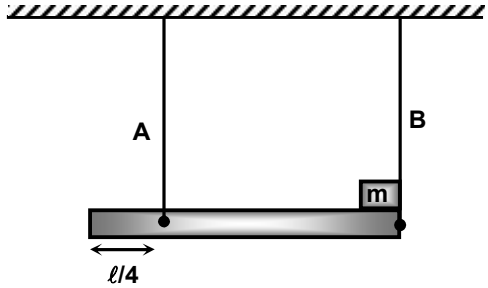
SECTION – I : PHYSICS

PART – A (Multiple Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

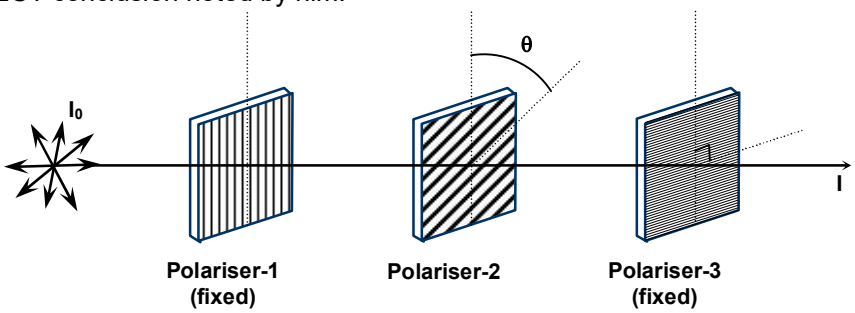
1. The width of a slit in a diffraction experiment is doubled (2x). The intensity of light at the central maximum
   (A) remains the same. (B) increases by 2 times.
   (C) increases by 4 times. (D) decreases by a factor of 2.

2. A uniform rod of mass 'm' and length 'ℓ' is held horizontally by two vertical strings 'A' and 'B' of negligible mass and a small block of mass 'm' is also placed on the top of the rod as shown in the figure. Then which of the following is/are correct.



   (A) The tension in the string 'A' immediately after the string 'B' is cut, is $\frac{4}{7}mg$.
   (B) The tension in the string 'A' immediately after the string 'B' is cut, is $\frac{2}{7}mg$.
   (C) The acceleration of centre of mass of the rod immediately after the string 'B' is cut, is $\frac{3}{7}g$.
   (D) The acceleration of centre of mass of the rod and the block immediately after the string 'B' is cut, is $\frac{5}{7}g$.

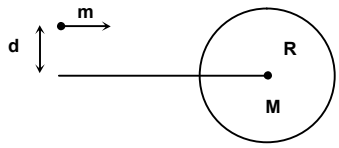
3. A student writes down four conclusions that he observes while performing a polarization experiment with 3 polarisers as shown in the figure. The leftmost and the rightmost polariser are fixed and crossed and the middle one can be rotated. The intensity of the unpolarised light is I_0. Identify the CORRECT conclusion noted by him.



   (A) If $\theta = 0°$ then $I = 0$ (B) If $\theta = 90°$ then $I = 0$
   (C) If $\theta = 45°$ then $I = \frac{I_0}{8}$ (D) If $\theta = 30°$ then $I = \frac{3I_0}{16}$

Space for rough work

4. A particle of mass m and velocity v_0 is fired at a solid cylinder of mass M and radius R. The cylinder is initially at rest and is mounted on a fixed horizontal axle that runs through the centre of mass. The line of motion of the particle is perpendicular to the axle and at a distance d, less than R, from the centre and the particle sticks to the surface of the cylinder, then



(A) Angular speed of the system just after the particle sticks is $\frac{2mv_0d}{R^2(M+2m)}$.

(B) Mechanical energy conserved.

(C) Angular speed of the system just after the particle sticks is $\frac{mv_0d}{R^2(M+2m)}$.

(D) Mechanical energy is not conserved.

5. During the propagation of electromagnetic waves in a medium:
(A) Electric energy density is equal to the magnetic energy density.
(B) Both electric and magnetic energy densities are zero.
(C) Electric energy density is double of the magnetic energy density.
(D) Electric energy density is half of the magnetic energy density.

6. Select the correct statement from the following
(A) Electromagnetic waves cannot travel in vacuum
(B) Electromagnetic waves are longitudinal waves
(C) Electromagnetic waves are produced by charges moving with uniform velocity.
(D) Electromagnetic waves carry both energy and momentum as they propagate through space.

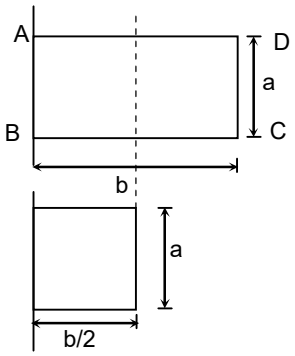
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PART – B
Integer Answer Type

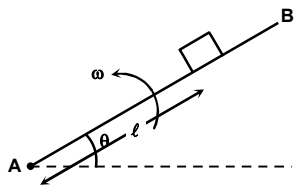
This section contains **06** questions. The answer to each of the questions is a **single digit integer**, ranging from **0 to 9.**

7. Unpolarised light passes through two polaroids, the axis of one is vertical and that of the other is at 45° to the vertical. Then, the intensity of the transmitted light is $\frac{I_0}{k}$. Find the value of k. (Where I_0 is intensity of unpolarized light)

8. Moment of inertia of a rectangular plate about one of its side AB is I_0. $\frac{I_0}{K}$ is the value of moment of inertia about same side if rectangular sheet is folded as shown in figure. Find the value of K.



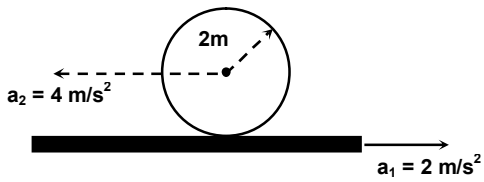
9. A plank is rotating in a vertical plane about one of its ends with a constant angular velocity $\omega = \sqrt{2}$ rad/s. A block of mass m = 2 kg is placed at a distance ℓ = 1 m from its end A (see figure) which is hinged. The block starts sliding down when the plank makes an angle $\theta = 30°$ with the horizontal. If coefficient of friction between the plank and the block is μ and given that $\mu^2 = k/25$. Find the value of k.



10. In an apparatus, the electric field was found to oscillate with an amplitude of 18 V/m. The magnitude of the oscillating magnetic field $k \times 10^{-8}$ T. Find the value of k.

11. Unpolarized light of intensity I passes through an ideal polarizer A. Another identical polarizer B is placed behind A. The intensity of light beyond B is found to be $\frac{I}{2}$. Now another identical polarizer C is placed between A and B. The intensity beyond B is now found to be $\frac{I}{8}$. The angle between polarizer A and C is 15K°. Find the value of K.

12. In the figure, a sphere of radius 2 metre is rolling without slipping on a plank. The accelerations of the sphere and the plank are indicated. Find the value of α (in rad/sec^2).



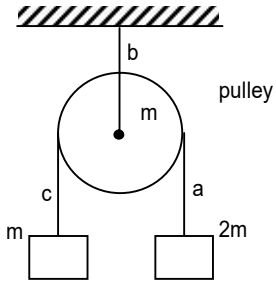
Space for rough work

PART – C
Integer Answer Type

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

13. A plane electromagnetic wave in a non magnetic dielectric medium is given by $\vec{E} = \vec{E}_0 \sin(4 \times 10^{-7} x - 50t)$ with distance being in meter and time in seconds. Refractive index of the medium is .....

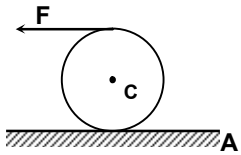
14. Mass of the pulley is m and radius is R. The acceleration of the block is $\frac{ng}{7}$. Find n. (Assume pulley as a disc and string does not slip over the pulley)



15. The light of wavelength 600 nm is incident normally on a slit of width 3mm. Calculate the linear width (in mm) of central maximum on a screen kept 3m away from the solid.

16. A parallel plate capacitor has circular plates, each of radius 5.0 cm. It is being charged so that electric field in the gap between its plates rises steadily at the rate of 10^{12} V/ms. Find the magnitude of displacement current (in ampere upto two decimal).

17. A massless string is wrapped around a solid cylinder having mass m and radius r. The cylinder is kept on a rough horizontal surface (coefficient of friction is μ). A constant force F is applied as shown in the figure. In case of pure rolling, the friction force acting on the bottom most point of the cylinder is F/k. Find k.



18. A beam of unpolarised light of intensity I_0 is passed through a polaroid A and then through another polaroid B which is oriented so that its principal plane makes an angle of 45° relative to that of A. The intensity of the emergent light is kI_0. Find the value of k.

Space for rough work

SECTION-II-CHEMISTRY

PART – A(I): (Single Correct Answer Type)

*This section contains **06 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.*

1. Identify the correct statements among the following: -
   (A) Among alkali metal, only Li does not form solid bicarbonates.
   (B) $KHCO_3$ is more soluble in water than $NaHCO_3$
   (C) Order of Melting point for alkali metals is : -
   Li > Na > K > Rb > Cs
   (D) Order of density of alkali metals is : -
   Li > Na > K > Rb > Cs

2. identify the correct orders among the following : -
   (A) Li < Mg (Atomic radii)
   (B) $Li^+ > Mg^{+2}$ (Ionic radii)
   (C) $LiC\ell > NaC\ell$ (Solubility in water)
   (D) In $H_3N \rightarrow BF_3$, B – F bond length is greater than the B – F bond length in BF_3

3. Which of the following molecules have $3C - 2e^-$ bond?
   (A) B_2H_6
   (B) $(BeH_2)_n$
   (C) $(BeC\ell_2)_n$
   (D) $Be(BH_4)_2$

Space for rough work

4. Identify correct orders among the following

(A) Na > Mg (2nd ionization energy)

(B) B < Ga < Aℓ < In < Tℓ (Atomic radii)

(C) $B^{+3} < A\ell^{+3} < Ga^{+3} < In^{+3}$ (Ionic radii)

(D) Catenation tendency : - C > Si > Ge > Sn

5. Correct order of Resonance energy is / are

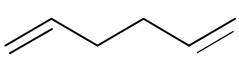
(A) acetate ion > Phenate ion

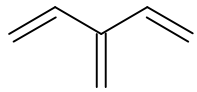
(B) Benzene > pyridine

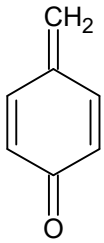
(C) Naphthalene > Benzene

(D) Pyrrole > Furan

6. Which among these is / are planar compound?

(A) 

(B) 

(C) 

(D) $H_2C{=}C{=}CH_2$

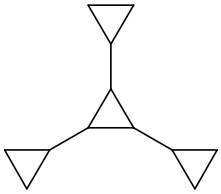
Space for rough work

PART-B (Numerical Based Type)

*This section contains **6** questions. The answer to each question is a single-digit integer, ranging from 0 to 9.*

7. How many three-Centre-two electron bonds (3C $-2e^-$) are present in $A\ell(BH_4)_3$?

8. How many geometrical isomers are possible for the following molecule?

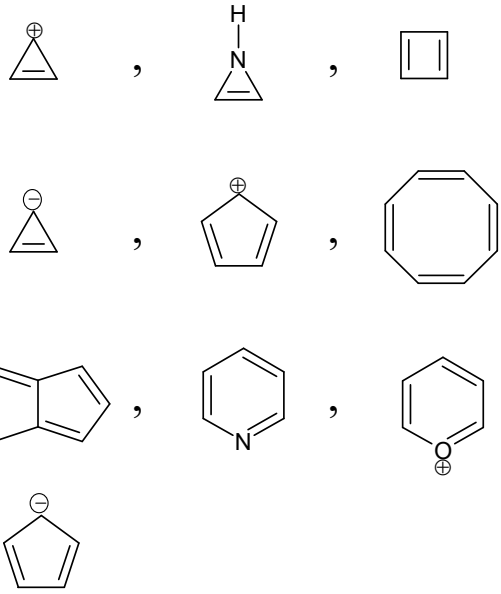


9. The ratio of σ and π - bonds in inorganic benzene is : -

10. How many molecules among the following will react with $NaHCO_3$?
Squaric acid, acetic acid, phenol, benzoic acid, Picric acid, ethanol, benzene sulphonic acid

Space for rough work

11. Of the given systems, how many are anti-aromatic systems?



12. Sum of Resonating structures of Naphthalene and Anthracene is : -

Space for rough work

PART-C (Numerical Decimal Based Type)

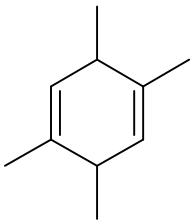
This section contains **6 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

13. Fullerene (C_{60}) contains x six membered rings and y five membered rings. What is the value of ratio x/y ?

14. Bond order of C – C bond in graphite is :-

15. Ratio of Number of correct and incorrect statements among the following : -
    (a) Borazine is isoelectronic with benzene.
    (b) Borazine is more reactive than benzene.
    (c) Both Borazine and benzene are planar molecules.
    (d) Both Borazine and benzene are aromatic molecules.
    (e) Resonance energy of borazine is greater than benzene.
    (f) In fullerene (C_{60}), five-membered rings are fused with only six-membered rings.
    (g) graphite is paramagnetic.

16. Consider the following molecule



Calculate ratio of $\frac{Total\ No.\ of\ possible\ stereoisomers}{No.\ of\ chiral\ carbon}$

17. 1 mole Gypsum $\xrightarrow{393\,K}$ product
    How many moles of water of crystallisation are present in the one mole product?

18. The No. of shared oxygen in $Sc_2\left[Si_2O_7\right]$ is x and the No. of shared oxygen in $BaTi\left[Si_3O_9\right]$ is y. The value of x/y will be :-

Space for rough work

SECTION – III : MATHEMATICS

PART – A: (Multi Correct Answer Type)

*This section contains **06 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.*

1. If the line ax + by + c = 0 is a normal to the curve xy = 1, then
   (A) $a > 0, b > 0$ (B) $a > 0, b < 0$
   (C) $a < 0, b > 0$ (D) $a < 0, b < 0$

2. Which one of the following curves cut the parabola $y^2 = 4ax$ at right angles?
   (A) $x^2 + y^2 = a^2$ (B) $y = e^{-x/2a}$
   (C) $y = ax$ (D) $x^2 = 4ay$

3. Let the parabola $y = x(c - x)$ and $y = x^2 + ax + b$ touch each other at the point (1, 0), then
   (A) $a + b + c = 0$ (B) $a + b = 2$
   (C) $b - c = 1$ (D) $a + c = -2$

4. If $I = \sum_{k=1}^{98} \frac{k+1}{x(x+1)}\,dx$, then
   (A) $I > \log_2 99$ (B) $I < \log_e 99$
   (C) $I < \frac{49}{50}$ (D) $I > \frac{49}{50}$

5. If f : R is a differentiable function such that $f'(x) > 2f(x)$ for all $x \in R$ and $f(0) = 1$, then
   (A) $f(x) > e^{2x}$ in $(0, \infty)$ (B) $f(x)$ is decreasing in $(0, \infty)$
   (C) $f(x)$ is increasing in $(0, \infty)$ (D) $f'(x) < e^{2x}$ in $(0, \infty)$

6. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1+\pi^x)}\,dx$, n = 0, 1, 2, ….., then
   (A) $I_n = I_{n+2}$ (B) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$
   (C) $\sum_{m=1}^{10} I_{2m} = 0$ (D) $I_n = I_{n+1}$

Space for rough work

PART – C: (Integer Answer Type)

*This section contains **06 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the OMR is to be bubbled.*

7. The value of $\int_0^1 4x^3\left(\frac{d^2}{dx^2}\left(1-x^2\right)^5\right)dx$ is equal to _____

8. If $\alpha=\int_0^1 e^{\left(9x+3\tan^{-1}x\right)}\left(\frac{12+9x^2}{1+x^2}\right)dx$ where $\tan^{-1}x$ takes only principal values, then the value of $\left(\log_e|1+\alpha|-\frac{3\pi}{4}\right)$ is equal to _____

9. If $\int_0^{100} f(x)\,dx=7$, then $\sum_{r=1}^{100}\left(\int_0^1 f(r-1+x)dx\right)=$ _____

10. If 'a', 'b' are two real numbers with a < b, then a real number 'c' can be found between 'a' and 'b' such that the value of $\frac{a^2+ab+b^2}{c^2}$ is equal to _____

11. Let P(x) be a real polynomial of least degree which has a local maximum at x = 1 and a local minimum at x = 3. If P(1) = 6 and P(3) = 2, then P′(0) is equal to _____

12. If $x=\text{cosec}\theta-\sin\theta$ and $y=\text{cosec}^3\theta-\sin^3\theta$ then the value of $\frac{\left(x^2+4\right)\left(\frac{dy}{dx}\right)^2}{\left(y^2+4\right)}$ is equal to _____

Space for rough work

PART – C: (Numerical Based Type)

*This section contains **06 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).*

13. The minimum value of $64\sec\theta + 27\operatorname{cosec}\theta$ when $\theta \in \left(0, \frac{\pi}{2}\right)$ is α, then the value of $\frac{\alpha}{100}$ is equal to _____

14. Let P(x) be a polynomial of degree 4, having extremum at x = 1 and x = 2, and $\lim\limits_{x\to 0}\left(1+\frac{P(x)}{x^2}\right) = 2$, then the value of $\frac{P(2)+1}{10}$ is equal to _____

15. The curve $y = ax^3 + bx^2 + cx + 5$ touches the x-axis at P(–2, 0) and cuts the y-axis at a point Q, where it's gradient is 3, then the value of |a + b + c| is equal to _____

16. If [x] denotes the greatest integer $\le x$ and $\alpha = \int_0^9 \left[\sqrt{x}\right] dx$, then the value of $\left(\frac{\alpha}{100}\right)$ is equal to _____

17. If $f(2) = \frac{\pi}{5}$ then the value of $\lim\limits_{x\to\frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(t)dt}{x^2 - \frac{\pi^2}{16}}$ is equal to _____

18. If $\alpha = \lim\limits_{n\to\infty}\left(\frac{(n+1)(n+2)...3n}{n^{2n}}\right)^{1/n}$ then the value of $\left(\frac{e^2\alpha}{10}\right)$ is equal to _____

Space for rough work

FIITJEE REVIEW TEST

PHYSICS, CHEMISTRY & MATHEMATICS

ANSWERS

PHYSICS (SECTION –I)

1. **C**	2. **A, C, D**	3. **A, B, C**	4. **A, D**
5. **A**	6. **D**	7. **4**	8. **4**
9. **3**	10. **6**	11. **3**	12. **3**
13. **2.40**	14. **2**	15. **1.20**	16. **0.07**
17. **3**	18. **0.25**		

CHEMISTRY (SECTION –II)

1. **A, B, C**	2. **A, B, C, D**	3. **A, B, D**	4. **A, B, C, D**
5. **A, B, C, D**	6. **B, C**	7. **6**	8. **2**
9. **4**	10. **5**	11. **5**	12. **7**
13. **1.67**	14. **1.33**	15. **2.50**	16. **1.50**
17. **0.5**	18. **0.33**		

MATHEMATICS (SECTION –III)

1. **B, C**	2. **B, D**	3. **A, C, D**	4. **B, D**
5. **A, C**	6. **A, B, C**	7. **2**	8. **9**
9. **7**	10. **3**	11. **9**	12. **9**
13. **1.25**	14. **0.1**	15. **1.75**	16. **0.13**
17. **1.6**	18. **2.7**		

HINTS & SOLUTIONS

PHYSICS (SECTION –I)

1. The amplitude is doubled, the intensity is quadrupled.

2. Just after cutting the string B, block looses the contact with the rod.
taking torque about P

$$mg\frac{\ell}{4}=\left(\frac{m\ell^2}{12}+\frac{m\ell^2}{16}\right)\alpha$$

$$\alpha=\frac{12}{7}\frac{g}{\ell},\ a_{cm}=\frac{\ell}{4}\alpha=\frac{3}{7}g$$

$mg - T = ma_{cm}$

$$T=\frac{4}{7}mg$$

3. Intensity of light after passing through polariser 1 is $I_0/2$.

Intensity of light after passing through polariser 2 is $\frac{I_0}{2}\cos^2\theta$.

Intensity of light after passing through polariser 3 is

$$I=\frac{I_0}{2}\cos^2\theta\cos^2(90-\theta)=\frac{I_0}{2}\sin^2\theta\cos^2\theta$$

So, for $\theta = 0$ or 90°, $I = 0$
for $\theta = 45°$, $I = I_0/8$

for $\theta = 30°$, $I=\frac{I_0}{2}\times\left(\frac{1}{2}\right)^2\left(\frac{\sqrt{3}}{2}\right)^2=\frac{3I_0}{32}$

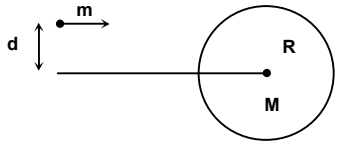
$\therefore$ d is incorrect.

4. $mv_0d = I\omega$

$$mv_0d=\left(\frac{MR^2}{2}+mR^2\right)\omega$$

$$\omega=\frac{2mv_0d}{R^2(M+2m)}$$

Energy is lost.



7. The first polaroid eliminates half the light, so the intensity is reduced by half, i.e., $I_1=\frac{I_0}{2}$. The light reaching the second polaroid is vertically polarised and so its intensity is reduced to $I_2=I_1(\cos 45°)^2=\frac{I_1}{2}$.

Thus, $I_2=\frac{I_0}{4}$, i.e. the transmitted light has an intensity one-fourth that of the original.

8. $I_0=Kmb^2$

$$I'=K.(m).\left(\frac{b}{2}\right)^2$$

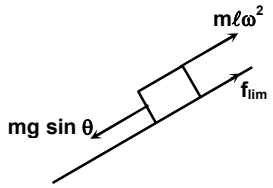
$$I'=\left(Kmb^2\right)\frac{1}{4}$$

$\Rightarrow \quad I' = \frac{I_0}{4}$.

9. $mg\sin\theta = f_{lim} + m l\omega^2$

$mg\sin\theta = \mu mg\cos\theta + m\omega^2\ell$

$\Rightarrow \quad \mu = \frac{g\sin\theta - \ell\omega^2}{g\cos\theta}$



10. $B_0 = \frac{E_0}{c} = \frac{18}{3\times10^8} = 6\times10^{-8}\,T$

11. $I_A = I/2$. But $I_B = I/2 \Rightarrow A \parallel B$

$\Rightarrow I/8 = \frac{I}{2}(\cos^2\theta)^2$

$\Rightarrow \cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45°$

12. $a_1 = R\alpha - a_2$

$2 = 2\alpha - 4 \Rightarrow \alpha = 3\ rad/s^2$.

13. $v = \frac{50}{4\times10^{-7}} \Rightarrow \mu = \frac{c}{v} = \frac{12}{5}$

15. Here, $\lambda = 600\ nm = 6\times10^{-7}\ m$

$a = 3mm = 3\times10^{-3}\ m$,

$D = 3m$

$\therefore$ width of central maximum $= \frac{2D\lambda}{a}$

$= \frac{2\times3\times6\times10^{-7}}{3\times10^{-3}} = 12\times10^{-4}\ m = 1.2\ mm$

16. $I_D = \varepsilon_0\frac{d\phi_E}{dt} = \varepsilon_0\pi r^2\frac{dE}{dt} = 8.85\times10^{-12}\times\pi\times(5\times10^{-2})^2\times10^{12} = 0.07\ A.$

18. If a beam of unpolarized light of intensity I_0 is passed through the two polaroids, the intensity of the emergent light is

$I = \frac{I_0}{2}\cos^2\theta$

$I = \frac{I_0}{2}\cos^2 45° = \frac{I_0}{4}$

CHEMISTRY (SECTION –II)

1. A, B, C

1. (A) Except Li, all other alkali metals form solid bicarbonates.

(B) Down the group is group No. 1, solubility of bicarbonates increases.

(C) Down the group in group No. 1, melting point decreases.

(D) Order of density for alkali metals is:- Li < K < Na < Rb < Cs

2. A, B, C, D

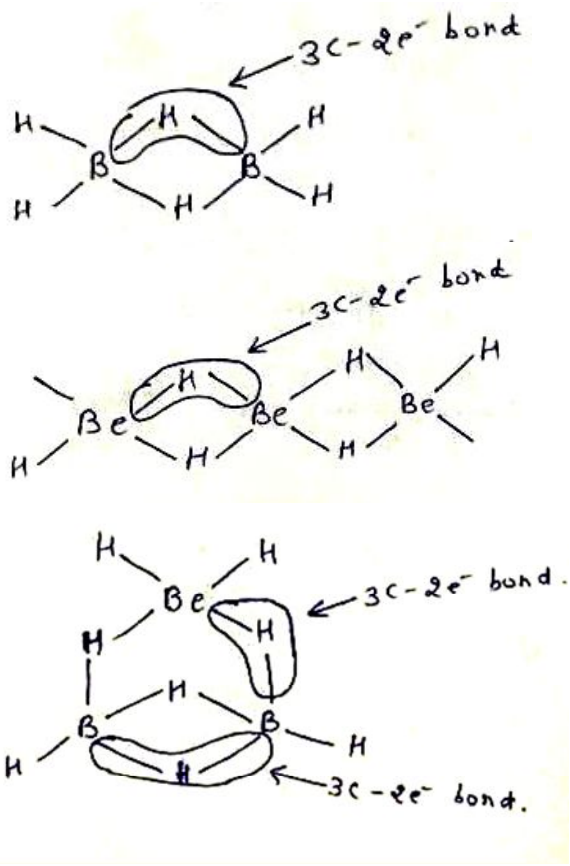
2. $\left.\begin{matrix}(a)\\(b)\end{matrix}\right\}\begin{matrix}factual\ question\\(\text{Refer } NCERT)\end{matrix}$

(C) LiCℓ is more soluble in water due to high hydration enthalpy of Li^+ ion.

(D) In BF_3, these is back bonding but in $NH_3 \rightarrow BF_3$, No back-bonding in B – F bond is possible.

3. A, B, D

3. B_2H_6, $(BeH_2)_n$ and $Be(BH_4)_2$ have 3C-2e$^-$ bonds.



4. A, B, C, D

4. (A) 2nd I.E. of Na > Mg because after loosing one electron Na acquires Noble gas configuration

(B) Trend of Atomic radii is irregular in gp. No. 13

(C) Ionic radii down the group increases.

(D) Down the group, catenation tendency decreases.

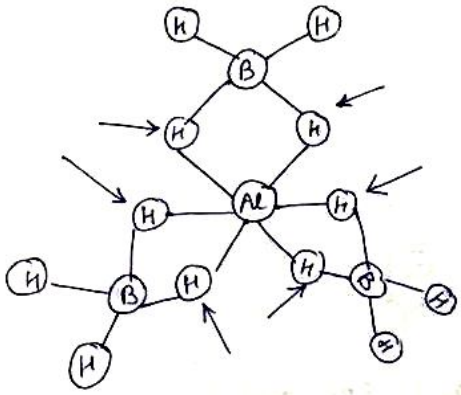
5. A, B, C, D

5. In acetate ion and benzene there is equivalent resonance. Naphthalene has greater No. of resonating structure as compared to benzene. Pyrole has more effective delocalization than furan due to less electro-negativity of N.

6. B, C

6. In option (a) and (d), all the atoms in the molecule are not in same plane. Therefore, they are non-planar molecules.

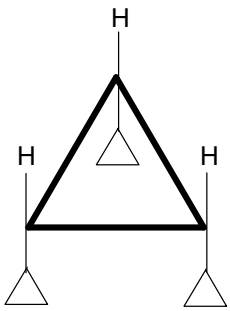
7. 6

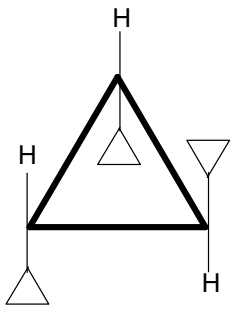
7. 

The arrows indicate 3C – $2e^-$ bonds. There are six 3C–$2e^-$ bonds in the given molecule.

8. 2

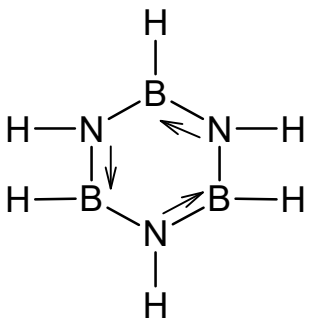
8. (1) (2)





(all cis)

9. 4

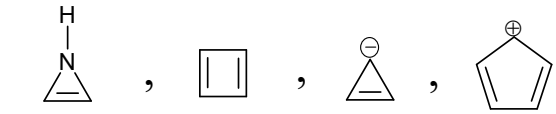
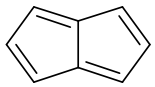
9. 

There are 12 σ and 3 π bonds in inorganic bezene.

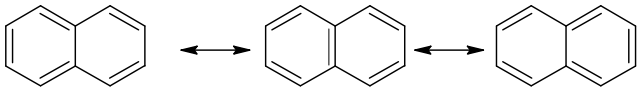
10. 5

10. Squaric acid, acetic acid, benzoic acid, picric acid and benzene sulphonic acid will react with $NaHCO_3$ and liberate effervescence of $CO_2(g)$ as these all molecules are more acidic than carbonic acid (H_2CO_3).

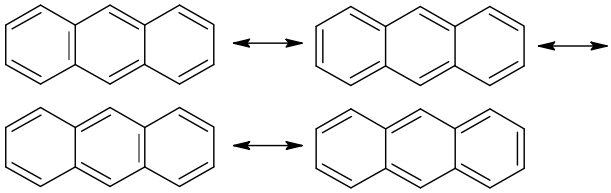
11. 5

11.   are anti - aromatic species.

12. 7

12. 

Resonating Structures of Naphthalene



Resonating Structures of anthracene

13. 1.67

13. Fullerene (C60) contains 20 six-mumbered rings and 12 Five – membered rings.

14. 1.33

14. Graphite has fused benzene rings just like Naphthalene

15. $\dfrac{Correct\ statement}{incorrect\ \ statement} = \dfrac{5}{2} = 2.50$

15. (a) borazine and benzene both contain same No. of electrons (48 e^-s)

(b) borazine is more reactive due to presence of polar bonds.

(c) In both molecular, all the atoms inside the ring are sp^2 hybridize. Both are planar and aromatic molecules.

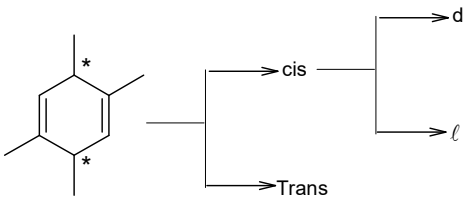
(e) Resonance energy of benzene is high because π e^-s are more effectively and symmetrically delocalized over the whole ring

(f) $\Rightarrow$ factual point [Refer to NCERT]

(g) graphite is diamagnetic.

Only (e) and (g) points are incorrect.

16. $\dfrac{3}{2} = 1.50$

16 

Trans – form is optically inactive due to presence of centre of symmetry.

$\therefore$ Total No. of stereo – isomers = 3 [cis – d, cis - ℓ and Trans]

Total No. of chiral carbon = 2

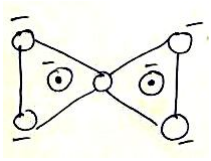
17. 0.5

17. $$CaSO_4 \cdot 2H_2O \xrightarrow{393\ K} CaSO_4 \cdot \frac{1}{2} H_2O$$

Gypsum (Plaster of Paris) [Product]

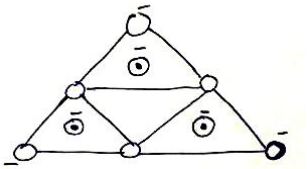
18. $\frac{1}{3} = 0.33$

18. Sc_2 [Si_2O_7] contains $Si_2O_7^{-6}$ unit (Pyro silicate unit) in which one O is shared b/n two SiO_4^{-4} units



$\left[Si_2O_7\right]^{-6}$

Ba Ti [Si_3 O_9] contains $\left[Si_3O_9\right]^{-6}$ unit which is a cyclic silicate in which three oxygen atoms are shared.



$\left[Si_3O_9\right]^{-6}$

MATHEMATICS (SECTION –III)

1. **B, C**

2. **B, D**

3. **A, C, D**
Slope of curve (1) at (1, 0)
= c – 2, slope of curve (2) at (1, 0)
= 2 + a $\Rightarrow m_1 = m_2$
Also (1, 0) lies on both the curves we get a = – 3, b = 2, c = 1

4. **B, D**

$$\sum_{k=1}^{98} \int_k^{k+1} \frac{1}{x+1}\,dx < \sum_{k=1}^{98} \int_k^{k+1} \frac{k+1}{x(x+1)}dx < \sum_{k=1}^{98} \int_k^{k+1} \frac{dx}{x}$$

$$\Rightarrow \sum_{k=1}^{98} \left(\log_e(k+2) - \log_e(k+1)\right) < I < \sum_{k=1}^{98} \left(\log_e(k+1) - \log_e k\right)$$

$$\Rightarrow \log_e 50 < I < \log_e 99$$

$$\Rightarrow \frac{49}{50} < I < \log_e 99\ .$$

5. **A, C**

$\frac{d}{dx}\left(e^{-2x}f(x)\right) > 0$ assume $g(x) = e^{-2x}f(x)$, also $g'(x) > 0\ \forall\ x \in R$ also $g(0) = 1$

$\Rightarrow$ $g(x) > g(0) \Rightarrow f(x) > e^{2x}\ \forall\ x \in (0, \infty)$

Also, $f'(x) > 2f(x) > 2e^{2x} > 2\ \forall\ (0, \infty)$

6. **A, B, C**

By property, we can prove $I_{n+2} = I_n$

Also, $I_1 = \pi$ and $I_2 = 0$

$\Rightarrow$ $I_1 = I_3 = I_5 = \ldots. = \pi$

And $I_2 = I_4 = I_6 = \ldots. = 0$

7. **2**

$$I = \left[4x^3 \frac{d}{dx}\left(1-x^2\right)^5\right]_0^1$$

$$-\int_0^1 12x^2 \frac{d}{dx}\left(1-x^2\right)^5 dx = 2$$

8. **9**

Put $9x + 3\tan^{-1}x = t$

We get $\alpha = \int_0^{9+\frac{3\pi}{4}} e^t\ dt$

$\Rightarrow$ $\log_e(1+\alpha) - \frac{3\pi}{4} = 9$

9. **7**

$$\sum_{r=1}^{100}\left(\int_0^1 f(r-1+x)dx\right) = \int_0^1 f(x)\ dx$$

$$+\int_1^2 f(x)\ dx + \ldots. \int_{99}^{100} f(x)dx = \int_0^{100} f(x)dx = 7$$

10. **3**

Applying LMVT for $f(x) = x^3\ \forall\ x \in [a, b]$, then

$$\frac{f(b)-f(a)}{b-a} = 3c^2 \Rightarrow \frac{b^2+ab+a^2}{c^2} = 3$$

11. **9**

$P'(x) = k(x-1)(x-3)$

$\Rightarrow$ $P(x) = k\left(\frac{x^3}{3} - 2x^2 + 3x\right) + c$

$\Rightarrow$ $c = 2$ and $k = 3$

$P'(0) = 3k = 9$

12. **9**

If $x = \text{cosec}\theta - \sin\theta$ and $y = \text{cosec}^n\theta - \sin^n\theta$ then,

$$\frac{\left(x^2+4\right)\left(\frac{dy}{dx}\right)^2}{\left(y^2+4\right)} = n^2$$

13. **1.25**

$f(\theta) = 64\sec\theta + 27\operatorname{cosec}\theta$

has minimum value at $\tan\theta = \frac{3}{4}$

$\Rightarrow \quad \alpha = 125$

14. **0.1**

By condition, P(x) comes

$P(x) = \frac{1}{4}x^4 - x^3 + x^2$

15. **1.75**

From condition, we get $a = -\frac{1}{2}, b = -\frac{3}{4}$ and $c = 3$

16. **0.13**

For $I_n = \int_0^{n^2} \left[\sqrt{x}\right] dx$

$= \frac{1}{6}(n-1)n(4n+1) \Rightarrow \alpha = 13$

$\Rightarrow \quad \frac{\alpha}{100} = 0.13$

17. **1.6**

Applying L hospital rule and Leibnitz formula we get the limit $= \frac{8}{\pi}f(2) \Rightarrow \frac{8}{\pi} \times \frac{\pi}{5} = 1.6$

18. **2.7**

By limit as a sum $\alpha = \frac{27}{e^2}$.